MILITARY SPECIFICATION SHEET

(D) ELECTRON TUBE, TRANSMITTING

TYPE 6293

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, pulse, beam power

Outline --- See figure 4

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	(D) 4	5	6	7	8	Cap
Element	---	k, g3, int sd	h	g2	k, g3, int sd	g1	k, g3, int sd	h	Base sleeve	a

ABSOLUTE-MAXIMUM RATINGS:

Modulator-rectangular-wave

Parameter:	Ef	(D) Ehk	Ebb	Ecc2	Ecc1	eb	ec1
Unit:	V	v	Vdc	Vdc	Vdc	v	v
Maximum:	6.9	±135	See figure 1 and note 1	500 (see figure 1 and note 1)	-300 (see figure 1 and note 2)	See note 3	+100 (see note 2)
Minimum:	5.7	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	---	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Modulator-rectangular-wave

Parameter:	ib	ic2	ic1	Pp	Pg2	Pg1	Pi	TE	(D) Alt
Unit:	a	a	a	W	W	W	W	°C	ft
Maximum:	See figure 2	0.75	0.5	See figure 1 and note 4	1.75	0.5	80	200	10,000
Minimum:	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1216	Base material insulating quality	Low-loss phenolic, zone 11 (min)	---	---	---	---	---	---
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = 300 Vdc; Ec2 = 200 Vdc; Ec1/Ib = 67 mAdc; t = 120 (see note 6)	0.65	II	Ic1	0	-3.0	µAdc
1256	Electrode current (1) (anode)	Eb = 300 Vdc; Ec2 = 200 Vdc; Ec1 = -33 Vdc	0.65	II	Ib	46	94	mAdc
1256	Electrode current (2) (anode)	Eb = 3,500 Vdc; Ec1 = -250 Vdc; Ec2 = 500 Vdc	0.65	II	Ib	---	1.0	mAdc
1256	Electrode current (screen)	Eb = 300 Vdc; Ec2 = 200 Vdc; Ec1 = -33 Vdc	0.65	II	Ic2	0	5.5	mAdc
---	Pulsed operation	Ebb = 3,500 Vdc; Ecc1 = -300 Vdc; Ecc2 = 500 Vdc; ec1 = +100 v; Rl = 1,000 ohms ±5 percent; C = 0.1 µF (see notes 5, 6, and 7)	0.65	II	ib	2.4	---	a
(D) ---	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Eb = 250 Vdc; Ec1/Ib = 10 mAdc; Rl = 2,000 ohms; Ec2 = 200 Vdc	---	---	Ep	---	750	mVac
1301	Heater current		---	---	If	1.175	1.325	A
1336	Heater-cathode leakage		---	---	Ihk	---	100	µAdc
1306	Transconductance	Eb = Ec2 = 200 Vdc; Ec1/Ib = 100 mAdc	---	---	Sm	5,600	8,400	µmhos
1331	Direct-interelectrode capacitance	Without shield, base sleeve tied to cathode	---	---	Cgp Cin Cout	--- 12.0 7.3	0.24 15.0 9.5	pF pF pF
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1111	Base pin solder depth		---	---	---	---	---	---
(D) 1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; pulsed operation; Ebb = 2,000 Vdc; Ecc2 = 500 Vdc; Ecc1 = -300 Vdc; R1 = 500 ohms ±5 percent; C = 2 μF; ec1 = +100 v (see note 8)	---	---	---	---	---	---
---	Life-test end point (500 hours)	Pulsed operation	---	---	ib	2.0	---	a
	Periodic-check test							
1131	Metallic base sleeve quality		---	---	---	---	---	---

NOTES:

1. For tube protection, it is essential that the dc resistance of the anode and screen supply be sufficiently large to limit the short circuit current to 0.5 amp, in each circuit.
2. The dc resistance of the grid circuit should not exceed 30,000 ohms. The instantaneous negative grid voltage due to transients shall not exceed -400 volts.
3. Instantaneous anode voltage shall never be greater than 115 percent of maximum dc anode supply voltage.
4. The anode dissipation shall be average over any period not exceeding 10,000 microseconds. If calculated, this anode dissipation shall be based on actual pulse shapes having finite rise and fall times.
5. A maximum of 5 arcs are permitted in the first 30 seconds of operation.
6. This test to be performed at the conclusion of the holding period.
7. (D) See circuit diagram figure 3. Use rectangular wave modulation on the grid. Adjust repetition rate to approximately 3,000 pulses per second for Ib = 9.0 mA minimum. Pulse width to be approximately 1 microsecond. Apply anode and screen voltages simultaneously.
8. See circuit diagram figure 3. Use rectangular wave modulation on the grid. Pulse width to be 27 to 30 microseconds. Repetition rate between 100 to 110 pulses per second.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI
Air Force - 17
DSA - ES

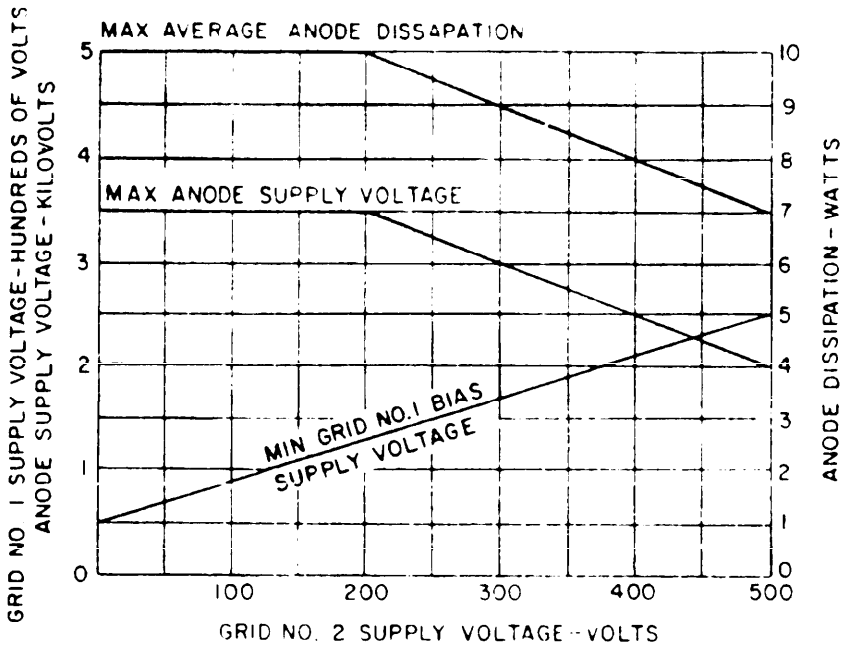
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

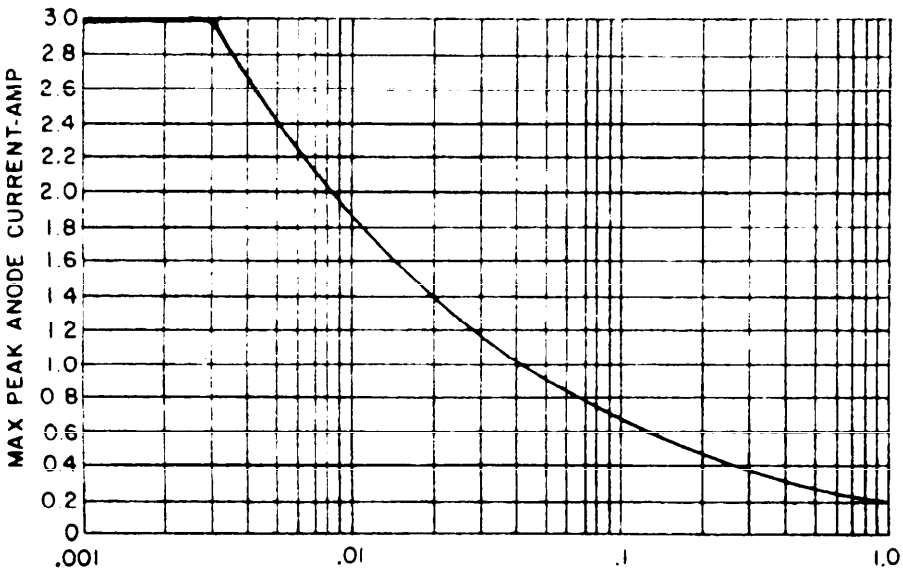
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(D)

FIGURE 1.



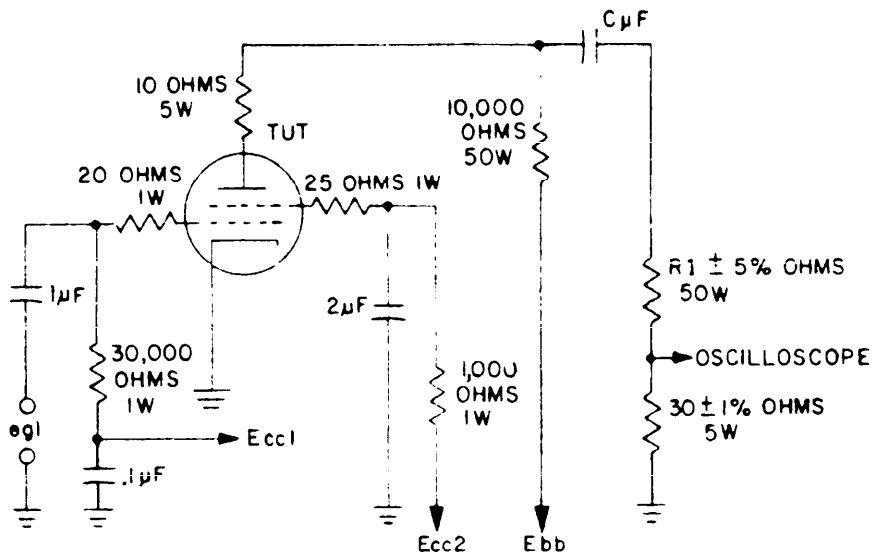
Duty factor for the 6293 is defined as the ratio of "on" time in microseconds to 10,000 microseconds.

"On" time is defined as the sum of the duration of all the individual pulses which occur during any 10,000 microsecond interval.

FIGURE 2.

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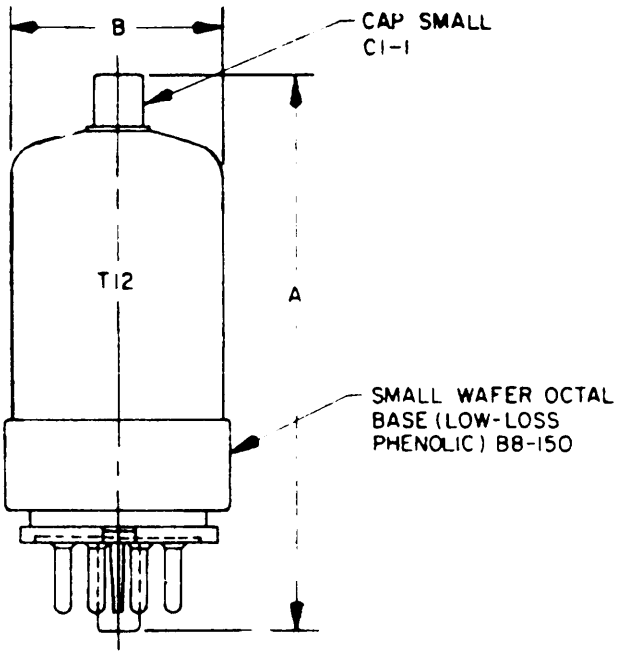
MODULATOR CIRCUIT DIAGRAM



NOTE: ALL 10, 20, 25, 30 AND R1 OHM RESISTORS SHALL BE NONINDUCTIVE

(D) FIGURE 2. Modulator circuit diagram.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		3.812(96.82)
B		1.562(39.67)

FIGURE 4. Outline drawing of electron tube type 6293.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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